

A large, faint, stylized smiley face is centered in the background. It is composed of several concentric blue circles and two horizontal blue ovals for eyes, all in a lighter shade of blue than the background.

challenge

a bit of fun

context **2023** meeting

The Sundqvist Smiley

This is how teens use potrace:

```
1 fill
2   lmt_potraced [ bytes =
3     "001111100
4     010000010
5     100000001
6     101101101
7     100000001
8     101000101
9     100111001
10    010000010
11    001111100",
12    polygon = false,
13    size = 1,
14  ] xysized (OverlayWidth-5mm,OverlayHeight-5mm)
15  withcolor "extracolor"
16  withtransparency (1,.3)
17  withpen pencircle scaled 0.1 ;
```

An alternate view

And this is how we modulate on that:

```
1 fill
2   lmt_potraced [ bytes =
3     "..11111.."
4     ".1.....1."
5     "1.....1"
6     "1.11.11.1"
7     "1.....1"
8     "1.1...1.1"
9     "1..111..1"
10    ".1.....1."
11    "..11111..",
12    polygon = false,
13    size = 1,
14  ] xysized (OverlayWidth-5mm,OverlayHeight-5mm)
15  withcolor "extracolor"
16  withtransparency (1,.3)
17  withpen pencircle scaled 0.1 ;
```

And we can go further, see next page.

Some more detail

```
1 draw image (
2   lmt_startpotraced [ bytes =
3     "..11111.."
4     ".1.....1."
5     "1.....1"
6     "1.22.22.1"
7     "1.....1"
8     "1.3...3.1"
9     "1..333..1"
10    ".1.....1."
11    "..11111.."
12  ] ;
13  fill lmt_potraced [ value = "1", size = 1 ]
14    withcolor "extracolor" withtransparency (1,.3)
15    withpen pencircle scaled 0.1 ;
16  fill lmt_potraced [ value = "3", size = 1 ]
17    withcolor "extracolor" withtransparency (1,.5)
18    withpen pencircle scaled 0.1 ;
19  fill lmt_potraced [ value = "2", size = 0 ]
20    withcolor "extracolor" withtransparency (1,.8)
21    withpen pencircle scaled 0.1 ;
22  lmt_stoppotraced ;
23  ) xysized (OverlayWidth-5mm,OverlayHeight-5mm) ;
```

This year X-mas card (1)

Here is how we use it ourselves: ©2023

```
1 path p ; p := lmt_potraced [  
2   stringname = "mybitmap",  
3   value      = "1",  
4   size       = 0,  
5 ] ;  
6 draw image (  
7   draw p withcolor "extracolor" withtransparency (1,.25) ;  
8   % fill p withcolor "darkred"   withtransparency (1,.3) ;  
9 ) xysized (OverlayWidth-5mm,OverlayHeight-5mm) ;
```

We also turned this into a game we played during the holidays and that later got properly materialized by Willy. We also made a service so that users could generate matching games in different complexity.

This year X-mas card (2)

```
1 \startluacode
2 local sin, cos = math.sin, math.cos
3
4 local N = 100
5
6 local function f(x,y)
7   local z = sin(x*x) - cos(y*y)
8   -- local z = sin(x) - cos(y)
9   if z > 0 then
10     return '1'
11   else
12     return '0'
13   end
14 end
15
16 potrace.setbitmap("mybitmap", potrace.contourplot(N,N,f))
17 \stopluacode
```